Deadbeat Control of a Three-Phase Inverter with an Output LC Filter

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***Abstract*—The discrete-time control of a three-phase inverter with an output LC filter is described based on space vectors. The mathematical model of the inverter-filter system is first obtained by using space vectors to represent three-phase quantities. Deadbeat control laws are derived for no-load and resistive-load cases. Then, a deadbeat control law is obtained for the case when the load draws current of any waveshape from the inverter-filter. It is shown that deadbeat control of output voltage can be achieved in two control steps. The manipulated variable, which is the voltage vector demanded from the inverter, is implemented using the space vector modulation technique. Simulation results for various operating conditions are presented.**

I. Introduction

THE deadbeat control of three-phase PWM inverters has been investigated by many researchers in recent years [1]–[4]. Feedback control of a three-phase inverter with filtered output is of particular importance for certain applications, such as uninterruptible power supplies, and for obtaining sinusoidal outputs under all loading conditions. Furthermore, such an inverter may have applications in variable-speed drives and in active filters. In [1] and [2], the deadbeat control of an inverter with an LC filter at its output is discussed. The inverter-filter system is modeled in an equivalent synchronously rotating two-phase frame after the application of a three-phase/two-phase transformation to the system equations. The result is that coupled equations are obtained in the equivalent two-phase frame, thus requiring decoupling transformations for controller design.

However, the mathematical modeling can be greatly simplified if space vectors are used in a stationary reference frame to represent three-phase quantities. This paper discusses the deadbeat control of a three-phase inverter with an output LC-filter based on space vectors. It is shown that deadbeat control of output voltage can be achieved in two control steps ideally, under no-load or with a purely resistive load. With other types of load, linear or nonlinear, it is shown that deadbeat control is still possible with an appropriate modification to the control law. In this case, the load is represented by the current it draws from the inverter-filter system, which may be sinusoidal or nonsinusoidal. Computer simulation results for the various cases studied are presented and discussed.

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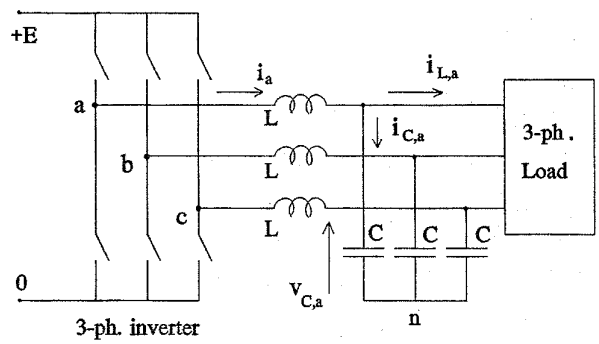

Fig. 1. A three-phase inverter with an LC filter at the output.

II. System Modeling

Fig. 1 shows a three-phase bridge inverter feeding a three-phase balanced load through a three-phase LC-filter. If the neutral point (n) of the filter and load are not connected to the midpoint of a split dc supply, the phase voltages of the load are related to the pole voltages of the inverter by

$$\begin{bmatrix} v_{an} \\ v_{bn} \\ v_{cn} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}\begin{bmatrix} v_{a0} \\ v_{b0} \\ v_{c0} \end{bmatrix}. \tag{1}$$

The phase voltages can be represented by the following space vector:

$$\mathbf{V} = v_{an} + \bar{\mathbf{a}} v_{bn} + \bar{\mathbf{a}}^2 v_{cn} \tag{2}$$

where $\bar{\mathbf{a}} = e^{j2\pi/3}$. Corresponding to each switching state of the inverter, eight voltage vectors will be generated, as shown in Fig. 2, which can be expressed as

$$\mathbf{V_m} = Ee^{j(m-1)\pi/3} \quad m = 1, \cdots, 6; \quad \mathbf{V}_7 = \mathbf{V}_8 = \mathbf{0}. \tag{3}$$

The zero vectors correspond to the switching states of the inverter in which either the upper or the lower switches are closed.

The following differential equations can be written for each phase of the system in Fig. 1

$$v_{pn} = L\frac{di_p}{dt} + v_{C,p} \tag{4}$$

$$i_p = C\frac{dv_{C,p}}{dt} + i_{L,p}. \tag{5}$$

In (4) and (5), p represents one of the phases a, b, or c. Defining the current space vector as

$$\mathbf{I} = i_a + \bar{\mathbf{a}} i_b + \bar{\mathbf{a}}^2 i_c. \tag{6}$$

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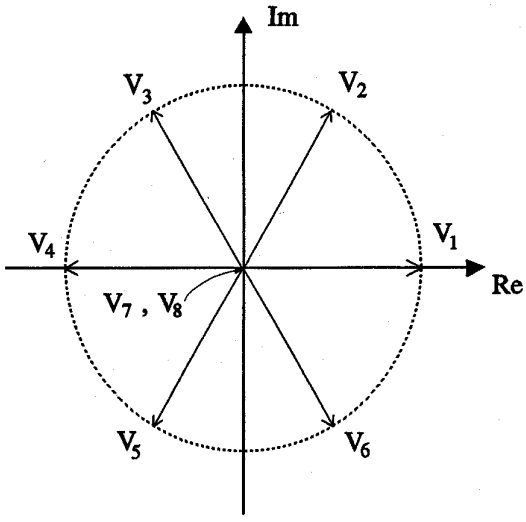


Fig. 2. Voltage vectors of the inverter.

Then (4) and (5) can be written as

$$\mathbf{V} = L\frac{d\mathbf{I}}{dt} + \mathbf{V}_C \tag{7}$$

$$\mathbf{I} = C\frac{d\mathbf{V}_C}{dt} + \mathbf{I}_L. \tag{8}$$

Equations (7) and (8) can be put into the following matrix form:

$$\frac{d\mathbf{x}}{dt} = F\mathbf{x} + G\mathbf{V} + H\mathbf{I}_L \tag{9}$$

where $\mathbf{x} = [\mathbf{V}_C \quad \mathbf{I}]^T$ is the state vector and T denotes transpose. In (9)

$$F = \begin{bmatrix} 0 & \frac{1}{C} \\ -\frac{1}{L} & 0 \end{bmatrix} \qquad G = \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix} \qquad H = \begin{bmatrix} -\frac{1}{C} \\ 0 \end{bmatrix}. \tag{10}$$

Assuming that the inputs $\mathbf{V}$ and $\mathbf{I}_L$ are constant between sampling instants kT and $(k+1)T$, (9) can be discretized to obtain the discrete-time state equation

$$\mathbf{x}(k+1) = e^{FT}\mathbf{x}(k) + (e^{FT} - U)F^{-1}[G\mathbf{V}(k) + H\mathbf{I}_L(k)]. \tag{11}$$

In (11), U is a 2×2 unit matrix, T is the sampling time and

$$A = e^{FT} = \begin{bmatrix} \cos(\omega_0 T) & \frac{1}{\omega_0 C}\sin(\omega_0 T) \\ -\frac{1}{\omega_0 L}\sin(\omega_0 T) & \cos(\omega_0 T) \end{bmatrix}, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\mathbf{b} = (e^{FT} - U)F^{-1}G = \begin{bmatrix} 1 - \cos(\omega_0 T) \\ \frac{1}{\omega_0 L}\sin(\omega_0 T) \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$\mathbf{d} = (e^{FT} - U)F^{-1}H = \begin{bmatrix} -\frac{1}{\omega_0 C}\sin(\omega_0 T) \\ 1 - \cos(\omega_0 T) \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}.$$

III. The Deadbeat Control Technique

Deadbeat control of the state vector in (11) will be obtained at no-load first. Hence equating to zero the load current $\mathbf{I}_L$, the state equation of the system in (11) reduces to

$$\mathbf{x}(k+1) = A\mathbf{x}(k) + \mathbf{b}\mathbf{V}(k). \tag{12}$$

Now, let

$$\mathbf{x}(k) = \mathbf{x}^*(k) + \Delta\mathbf{x}(k) \tag{13}$$

$$\mathbf{V}(k) = \mathbf{V}^*(k) + \Delta\mathbf{V}(k) \tag{14}$$

where $\mathbf{x}^*(k)$ and $\mathbf{V}^*(k)$ are reference vectors for the state and voltage vectors, respectively, at the kth sampling instant. Substituting (13) and (14) in (12) gives

$$\mathbf{x}^*(k+1) + \Delta\mathbf{x}(k+1) = A\mathbf{x}^*(k) + A\Delta\mathbf{x}(k) + \mathbf{b}(\mathbf{V}^*(k) + \Delta\mathbf{V}(k)). \tag{15}$$

If we define

$$\mathbf{x}^*(k+1) = A\mathbf{x}^*(k) + \mathbf{b}\mathbf{V}^*(k) \tag{16}$$

then (15) reduces to

$$\Delta\mathbf{x}(k+1) = A\Delta\mathbf{x}(k) + \mathbf{b}\Delta\mathbf{V}(k). \tag{17}$$

In (17) if the following state feedback is used

$$\Delta\mathbf{V}(k) = -K\Delta\mathbf{x}(k) \quad K = [k_1 \quad k_2] \tag{18}$$

then, the closed-loop state equation becomes

$$\Delta\mathbf{x}(k+1) = (A - \mathbf{b}K)\Delta\mathbf{x}(k). \tag{19}$$

The closed-loop system matrix is

$$A - \mathbf{b}K = \begin{bmatrix} a_{11} - b_1 k_1 & a_{12} - b_1 k_2 \\ a_{21} - b_2 k_1 & a_{22} - b_2 k_2 \end{bmatrix}.$$

The characteristic equation of the system in (19) is

$$|sU - (A - \mathbf{b}K)| = s^2 + \alpha s + \beta \tag{20}$$

where

$$\alpha = b_1 k_1 + b_2 k_2 - 2a, \qquad a = a_{11} = \cos(\omega_0 T)$$
$$\beta = 1 + b_1 k_1 - b_2 k_2.$$

Deadbeat control will be obtained if k_1 and k_2 are chosen so that $\alpha = \beta = 0$. The solution for k_1 and k_2 is

$$k_1 = \frac{2a - 1}{2b_1} \qquad k_2 = \frac{2a + 1}{2b_2}. \tag{21}$$

With the gains as in (21), the closed-loop system matrix becomes

$$A_c = A - \mathbf{b}K = \begin{bmatrix} \frac{1}{2} & \frac{1}{2}X_L \tan\left(\frac{\theta_s}{2}\right) \\ -\frac{1}{2X_L}\cot\left(\frac{\theta_s}{2}\right) & -\frac{1}{2} \end{bmatrix}. \tag{22}$$

In (22) $\theta_s = \omega_0 T$ and $X_L = \omega_0 L$. It can be easily verified that $A_c^2 = \mathbf{0}$. This implies that for an arbitrary error $\Delta\mathbf{x}(k)$ at step k

$$\Delta\mathbf{x}(k+2) = A_c^2 \cdot \Delta\mathbf{x}(k) = \mathbf{0}.$$

IV. The Reference Vector

In order to implement the control law in (14) and (18), the reference for the inverter voltage vector ($\mathbf{V}^*(k)$) must be evaluated. Now, since the controlled variable is the capacitor voltage vector $\mathbf{V}_C$,the reference for this vector is to be specified. Then, the references for the inverter voltage and current vectors, $\mathbf{V}^*$ and $\mathbf{I}^*$, can be obtained from (16). For sinusoidal variations

$$\begin{aligned}\mathbf{V}_C^*(k+1) &= \mathbf{V}_{\mathrm{cr}}e^{j(k+1)\omega T}\\ &= e^{j\omega T}\mathbf{V}_C^*(k) = q\mathbf{V}_C^*(k) \qquad (23)\\ \mathbf{I}^*(k+1) &= q\mathbf{I}^*(k). \qquad (24)\end{aligned}$$

Substituting (23) and (24) in (16) and solving yields

$$\mathbf{V}^*(k) = \mathbf{V}_C^*(k)\frac{q^2-2aq+1}{(1-a)(1+q)} \qquad (25)$$

$$\mathbf{I}^*(k) = \mathbf{V}_C^*(k)\frac{b_2(q-1)}{(1-a)(1+q)}. \qquad (26)$$

V. Deadbeat Control with Resistive Load

When the load of the inverter is purely resistive, with per phase resistance R_0, then the state equations of the inverter-load system can be written as

$$\frac{d\mathbf{x}}{dt} = F'\mathbf{x} + G'\mathbf{V} \qquad (27)$$

where

$$F' = \begin{bmatrix} \frac{-1}{R_0C} & \frac{1}{C} \\ \frac{-1}{L} & 0 \end{bmatrix} \qquad G' = \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix}.$$

Then the discrete-time state equations for the system can be written as

$$\mathbf{x}(k+1) = A'\mathbf{x}(k) + \mathbf{b}'\mathbf{V}(k). \qquad (28)$$

Assuming that the response of the system in (27) is underdamped, the elements of the matrices $\mathbf{A}'$ and $\mathbf{b}'$ are

$$\begin{aligned} a'_{11} &= -\gamma e^{-\xi\omega_0 T}\sin(\omega_d T - \phi)\\ a'_{12} &= \frac{1}{\omega_0 C}\gamma e^{-\xi\omega_0 T}\sin(\omega_d T)\\ a'_{21} &= -\frac{1}{\omega_0 L}\gamma e^{-\xi\omega_0 T}\sin(\omega_d T)\\ a'_{22} &= \gamma e^{-\xi\omega_0 T}\sin(\omega_d T + \phi) \qquad (29)\\ b'_1 &= 1 - a'_{11} - \frac{a'_{12}}{R_0}\\ b'_2 &= -a'_{21} + \frac{1-a'_{22}}{R_0}. \end{aligned}$$

In (29), w_0 is the undamped natural frequency of the LC filter defined before and

$$\xi = \frac{1}{2R_0}\sqrt{\frac{L}{C}} \qquad \gamma = \frac{1}{\sqrt{1-\xi^2}}$$

$$\omega_d = \sqrt{1-\xi^2}\omega_0 \qquad \phi = \tan^{-1}\left(\frac{\sqrt{1-\xi^2}}{\xi}\right).$$

Following the same procedure for the no-load case, one obtains the following gains for the deadbeat controller:

$$k'_1 = \frac{1}{D}[(a'_{21}b'_1 - a'_{11}b'_2)(a'_{11}+a'_{22}) + b'_2] \qquad (30a)$$

$$k'_2 = \frac{1}{D}[(a'_{12}b'_2 - a'_{22}b'_1)(a'_{11}+a'_{22}) + b'_1] \qquad (30b)$$

where

$$D = (a'_{22} - a'_{11})b'_1b'_2 + a'_{21}{b'_1}^2 - a'_{12}{b'_2}^2.$$

The inverter voltage and current reference vectors for the resistive load case must also be modified as follows:

$$\mathbf{V}^*(k) = \frac{1}{Q}[(q-a'_{11})(q-a'_{22}) - a'_{12}a'_{21}]\mathbf{V}_C^*(k) \qquad (31a)$$

$$\mathbf{I}^*(k) = \frac{1}{Q}[b'_2(q-a'_{11}) + b'_1a'_{21}]\mathbf{V}_C^*(k) \qquad (31b)$$

where

$$Q = b'_1(q-a'_{22}) + b'_2a'_{12}.$$

VI. Deadbeat Control with Any Load

When the load of the inverter-filter system is such that it draws current of any waveshape (sinusoidal or nonsinusoidal) from the filter, then the state equations of the system can be written as

$$\mathbf{x}(k+1) = A\mathbf{x}(k) + \mathbf{b}\mathbf{V}(k) + \mathbf{d}\mathbf{I}_L(k) \qquad (32)$$

where the matrices $\mathbf{A}$, $\mathbf{b}$ and $\mathbf{d}$ are as defined in (11). With the definitions in (13), (14), and (16), we have

$$\Delta\mathbf{x}(k+1) = A\Delta\mathbf{x}(k) + \mathbf{b}\Delta\mathbf{V}(k) + \mathbf{d}\mathbf{I}_L(k). \qquad (33)$$

Let the control law in (18) be modified as

$$\Delta\mathbf{V}(k) = -K\Delta\mathbf{x}(k) + \Delta\mathbf{V}^l(k). \qquad (34)$$

Then, the closed-loop system state equation is

$$\Delta\mathbf{x}(k+1) = A_c\Delta\mathbf{x}(k) + \mathbf{b}\Delta\mathbf{V}^l(k) + \mathbf{d}\mathbf{I}_L(k). \qquad (35)$$

If (35) is written for the next control step, then

$$\Delta\mathbf{x}(k+2) = A_c\Delta\mathbf{x}(k+1) + \mathbf{b}\Delta\mathbf{V}^l(k+1) + \mathbf{d}\mathbf{I}_L(k+1). \qquad (36)$$

Substituting (35) in (36) gives

$$\begin{aligned}\Delta\mathbf{x}(k+2) &= A_c\mathbf{b}\Delta\mathbf{V}^l(k) + A_c\mathbf{d}\mathbf{I}_L(k)\\ &\quad + \mathbf{b}\Delta\mathbf{V}^l(k+1) + \mathbf{d}\mathbf{I}_L(k+1). \qquad (37)\end{aligned}$$

In (37) the fact that $A_c^2 = \mathbf{0}$ has been used. Also, it can be easily shown that

$$A_c\mathbf{b} = \begin{bmatrix} 2\sin^2\left(\frac{\theta_s}{2}\right) \\ -\frac{1}{X_L}\sin(\theta_s) \end{bmatrix} = \begin{bmatrix} b_1 \\ -b_2 \end{bmatrix} \qquad (38)$$

$$A_c\mathbf{d} = \begin{bmatrix} -X_C\tan\left(\frac{\theta_s}{2}\right)\cos(\theta_s) \\ \cos(\theta_s) \end{bmatrix}. \qquad (39)$$

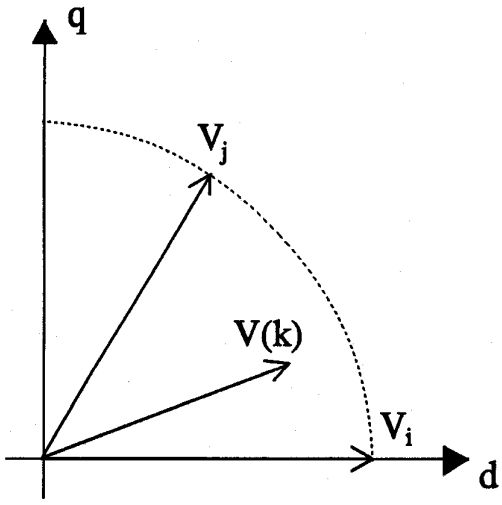


Fig. 3. Voltage vector $\mathbf{V}(k)$ in the sector between $\mathbf{V}_i$ and $\mathbf{V}_j$.

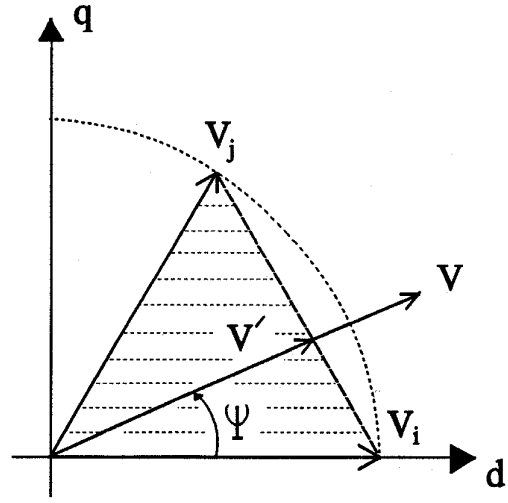


Fig. 4. Region of synthesizable voltage vectors (shaded area).

In (39) $X_C = \frac{1}{\omega_0 C} = X_L$. In (37) we require that the change in the capacitor voltage $\Delta\mathbf{V}_C$ be zero for deadbeat control. Using (38) and (39)

$$\begin{aligned}\Delta\mathbf{V}_C(k+2) &= b_1(\Delta\mathbf{V}^l(k) + \Delta\mathbf{V}^l(k+1)) \\ &\quad + \left(-X_C \tan\left(\frac{\theta_s}{2}\right)\cos(\theta_s)\right)\mathbf{I}_L(k) \\ &\quad - (X_C \sin(\theta_s))\mathbf{I}_L(k+1) = 0. \end{aligned} \tag{40}$$

The correction term in the manipulated inverter voltage vector is thus updated according to

$$\begin{aligned}\Delta\mathbf{V}^l(k+1) &= -\Delta\mathbf{V}^l(k) + (X_C \cot(\theta_s))\mathbf{I}_L(k) \\ &\quad + \left(X_C \cot\left(\frac{\theta_s}{2}\right)\right)\mathbf{I}_L(k+1). \end{aligned} \tag{41}$$

VII. PWM Approximation of the Voltage Vector $\mathbf{V}(k)$

The inverter voltage vector $\mathbf{V}(k)$ demanded by the control algorithm in (14) is an arbitrary complex quantity and hence cannot be generated by the inverter. Therefore, it must be approximated by the available voltage vectors $\mathbf{V}_m$, $m = 1, \cdots, 8$. One approach is to transform the d- and q-axis components, $\mathbf{V}_d(k)$ and $\mathbf{V}_q(k)$ to three-phase and generate the switching functions by comparison with a triangular carrier signal synchronized with the control sampling. A second approach is to obtain directly the inverter switching functions using the space vector modulation technique [5]–[7], which is briefly described below. This approach is simpler to implement than the first one and is the approach used here.

Fig. 3 shows a voltage vector $\mathbf{V}(k)$ in the sector defined by $\mathbf{V}_i$ and $\mathbf{V}_j$, which are two consecutive nonzero vectors of the inverter ($j = i+1$). The vector $\mathbf{V}(k)$ can be synthesized with the inverter voltage vectors as

$$T\mathbf{V}(k) = T_1\mathbf{V}_i + T_2\mathbf{V}_j. \tag{42}$$

In (42) T_1 and T_2 are the durations of the inverter voltage vectors $\mathbf{V}_i$ and $\mathbf{V}_j$, respectively, which can be solved for from (42) as

$$T_1 = \frac{1}{\Delta_{ij}}[\mathbf{V}_{jq}\mathbf{V}_d - \mathbf{V}_{jd}\mathbf{V}_q]T \tag{43}$$

$$T_2 = \frac{1}{\Delta_{ij}}[\mathbf{V}_{id}\mathbf{V}_q - \mathbf{V}_{iq}\mathbf{V}_d]T \tag{44}$$

where

$$\Delta_{ij} = \mathbf{V}_{jq}\mathbf{V}_{id} - \mathbf{V}_{jd}\mathbf{V}_{iq}.$$

The duration of the zero vector ($\mathbf{V}_7$ or $\mathbf{V}_8$) is then

$$T_0 = T - T_1 - T_2 \tag{45}$$

which is usually split into two [7]. This means that the zero vector is applied for equal durations of $T_0/2$ at the beginning and at the end of a sampling period. The case $T_1 + T_2 > T$ means that $\mathbf{V}(k)$ cannot be synthesized with the available inverter voltage vectors in the sampling interval T (overmodulation). It can be easily shown that

$$T_1 + T_2 = \frac{2}{\sqrt{3}}\left(\frac{\mathbf{V}}{E}\right)\sin\left(\frac{2\pi}{3} - \psi\right)T \tag{46}$$

where ψ is defined in Fig. 4. The condition that $\mathbf{V}(k)$ can be synthesized is then

$$\mathbf{V}\sin\left(\frac{2\pi}{3} - \psi\right) \le \frac{\sqrt{3}}{2}E. \tag{47}$$

Equation (47) defines a region of synthesizable voltage vectors in the sector between $\mathbf{V}_i$ and $\mathbf{V}_j$, as shown in Fig. 4. Using (46) it can be shown that the scaling of the durations T_1 and T_2 given in (48) scales the magnitude of the vector $\mathbf{V}$ to a point on the boundary in Fig. 4

$$T_1' = \frac{T_1}{T_1 + T_2}T \qquad T_2' = \frac{T_1}{T_1 + T_2}T. \tag{48}$$

It should be pointed out that, in case the inverter switches can operate at frequencies much higher than the control

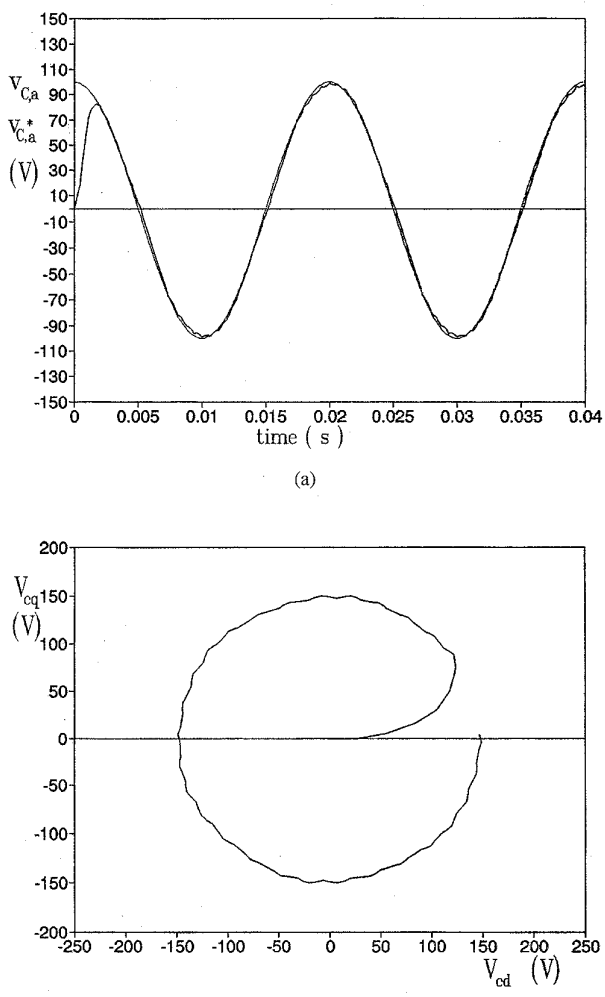


Fig. 5. (a) Phase A capacitor voltage ($v_{C,a}$) and its sinusoidal reference at no-load. (b) Corresponding capacitor voltage vector.

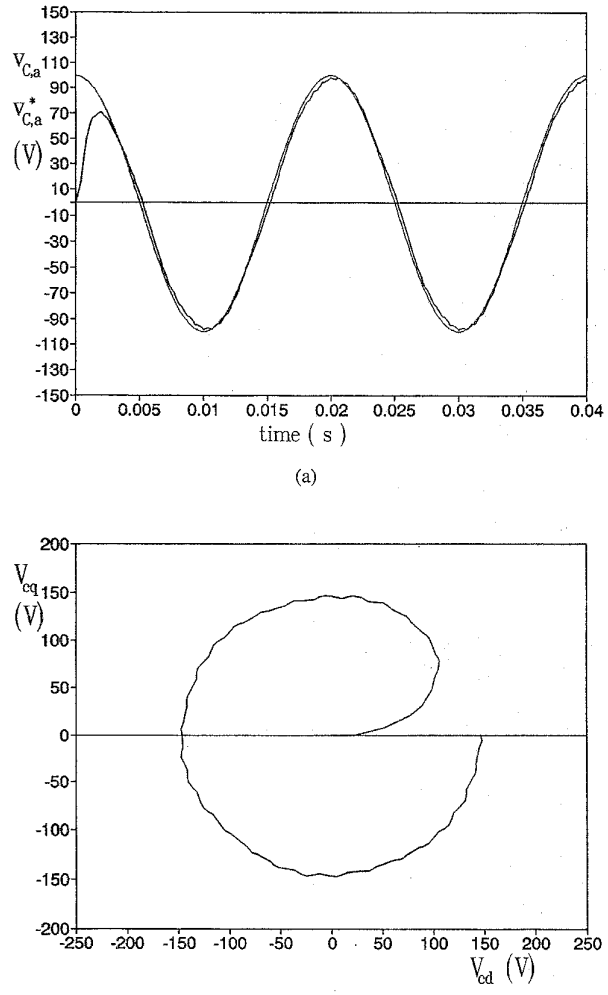


Fig. 6. (a) Phase A capacitor voltage ($v_{C,a}$) and its reference with resistive load ($R_0 = 4\ \Omega$). (b) Corresponding capacitor voltage vector.

sampling frequency, the sequence of the inverter vectors as determined above can be applied more than once in a sampling interval. In that case, the durations given above should be scaled accordingly.

VIII. Computer Simulation Results

The proposed deadbeat control strategy has been simulated for various operating conditions. In computing the responses of the output voltage, the following assumptions have been made:

1) There is no time delay between sampling and the implementation of control, caused by the control calculations. Such a delay would arise in a practical application.
2) The load current is constant in a sampling interval. This assumption is valid if the control sampling frequency is much higher than the operating frequency of the inverter. This assumption has been made to facilitate the computations for any load current waveshape.

The effect the filter parameters and the sampling time would have on the performance of the system is a problem that needs a separate study. Here, representative values have been chosen which are: $L = 0.91171$ mH, $C = 1000\ \mu$F, and $T = 1/1500$ s, resulting in the sampling angle $\theta_s = 40°$ for all the cases studied. The dc input voltage of the inverter is $E = 250$ V.

Computation of the responses are started by initializing all variables to zero and then applying the reference voltage vector as a step at $t = 0$.

Fig. 5(a) shows phase A capacitor voltage ($v_{C,a}$) and its sinusoidal reference ($v^*_{C,a}$) at no load. In Fig. 5(b), the capacitor voltage vector is shown for this case. It can be observed that the capacitor voltage is nearly sinusoidal with negligible distortion. Fig. 6(a) shows the capacitor voltage ($v_{C,a}$) with a resistive load of $R_0 = 4\ \Omega$ per phase-Y, and Fig. 6(b) shows the corresponding voltage vector. With this load resistance, the damping ratio in (29) is $\xi = 0.12$. Note that the response of load voltage is similar to the no-load case, with slightly increased deviation from the reference wave. A case of particular importance arises when the load resistance deviates

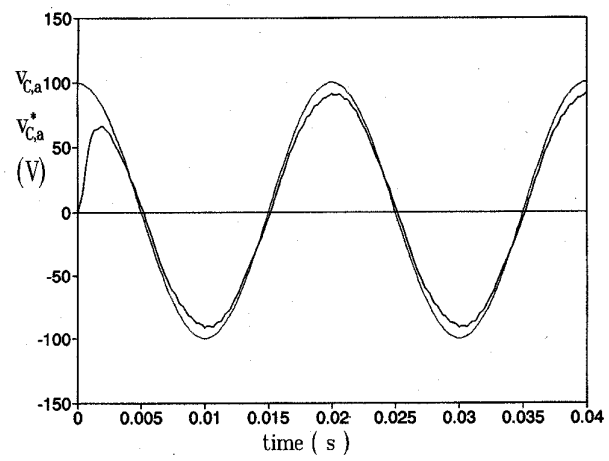
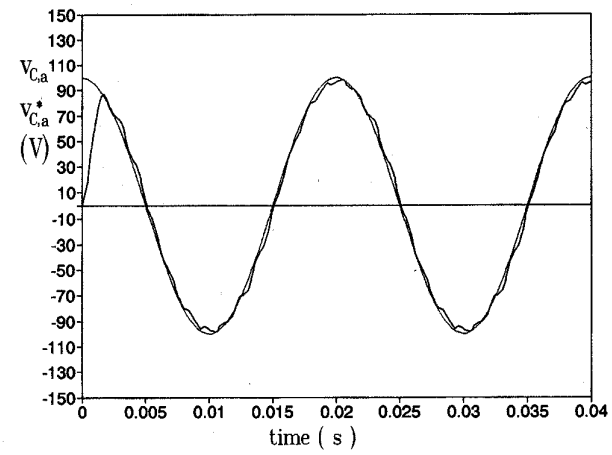
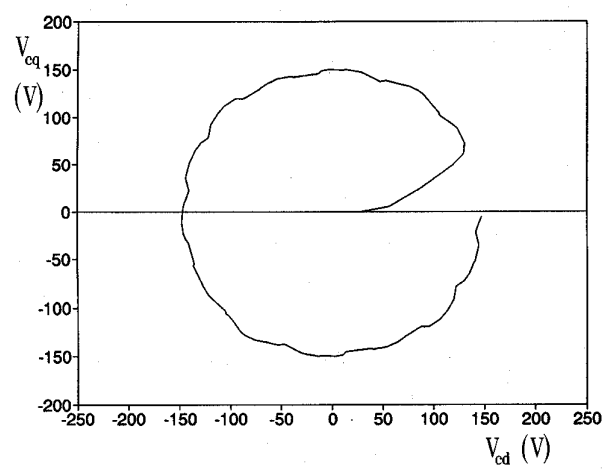


Fig. 7. Phase A capacitor voltage with load resistance $R_0 = 3\ \Omega$.



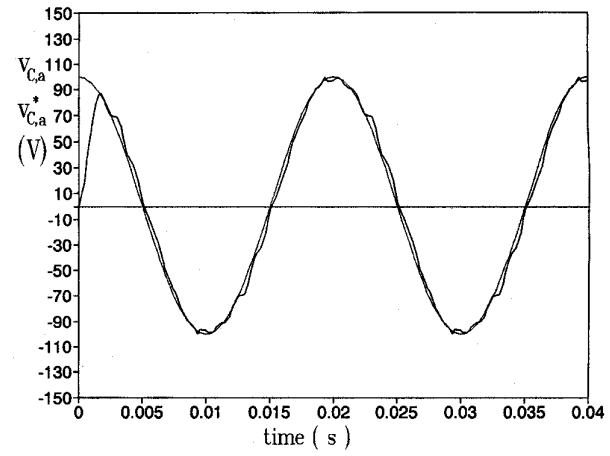
(a)



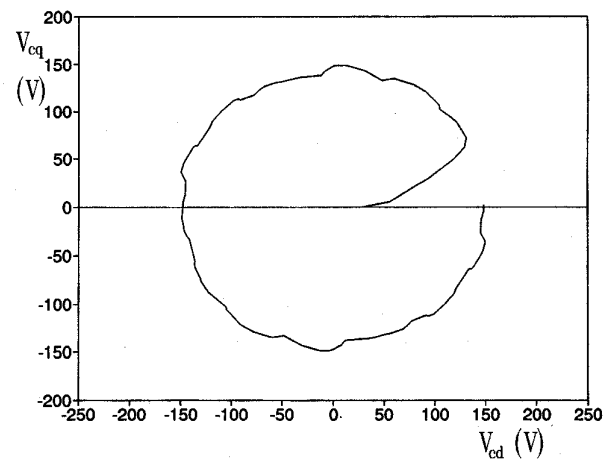
(b)

Fig. 8. (a) Phase A capacitor voltage ($v_{C,a}$) and its reference with 20 A (amplitude) load current. (b) Corresponding capacitor voltage vector.

from the nominal value used for gain and reference vector calculations. In this case the poles of the system shift away from origin. Simulations show that the system stays stable for



(a)



(b)

Fig. 9. (a) Phase A capacitor voltage and its reference with a three-phase rectifier load (3-ph quasi-square-wave load current). (b) Corresponding capacitor voltage vector.

all R_0 in the range

$$R_0 > \frac{1}{2}\sqrt{L/C}. \tag{49}$$

Fig. 7 shows the load voltage for $R_0 = 3\ \Omega$. The error between controlled and reference voltages is due to the fact that the reference vectors required for the load $R_0 = 3\ \Omega$ are different from the ones calculated at the nominal value $R_0 = 4\ \Omega$.

Fig. 8(a) shows the capacitor voltage ($v_{C,a}$) with a sinusoidal load current of 20 A amplitude and lagging behind $v_{C,a}$ by 45°. Although the distortion in $v_{C,a}$ increases, it is within reasonable limits and can be improved by increasing the sampling rate. Fig. 8(b) shows the voltage vector for this case.

The responses of the control system with a nonlinear load are shown in Fig. 9. The load here is a three-phase bridge rectifier with a constant output current of 18 A, demanding a quasi-square-wave current from the filter, as shown in Fig. 10. The space phasor for this load current is

$$\mathbf{I}_L(t) = \frac{3\sqrt{3}}{\pi} I_d \left(e^{j(wt+\phi_L)} - \frac{1}{5} e^{-j5(wt+\phi_L)} + \frac{1}{7} e^{j7(wt+\phi_L)} - \cdots \right) \tag{50}$$

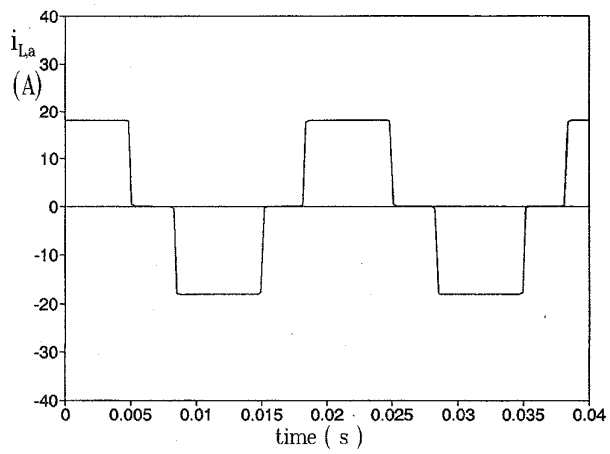


Fig. 10. The quasi-square-wave load current used for the responses in Fig. 9.

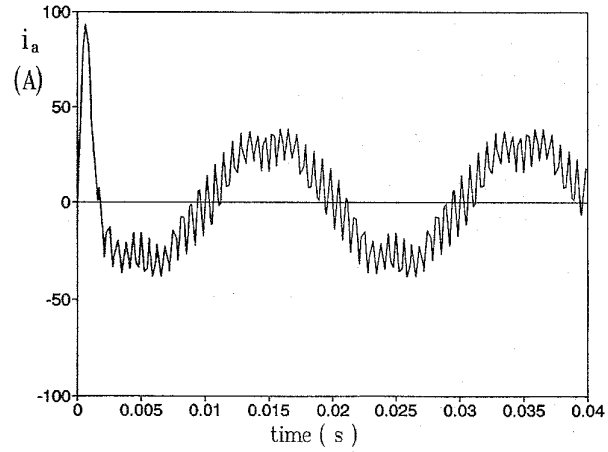
where I_d is the dc load current and ϕ_L is the phase shift of the fundamental with respect to the voltage reference V_C^*. Again, it can be observed from the responses that the increase in distortion with respect to the no-load case is not very significant.

Fig. 11(a)–(c) shows the currents supplied by the inverter in the no-load, resistive load and lagging power factor load cases of Figs. 5(a), 6(a), and 8(a), respectively. Note that the ripple amplitude in the inverter current for the lagging power factor load is much larger than that for the no-load case or the resistive load.

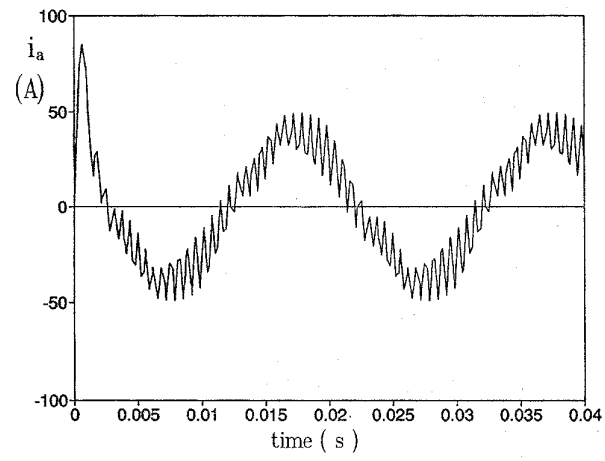
Fig. 12(a) shows the response of phase A capacitor voltage to a variable frequency reference. Here, the frequency of the reference is changed from 0.4 Hz to 50 Hz in steps of 0.2 Hz per sampling period. The amplitude of the capacitor voltage reference ($V_{\rm cr}$) is increased linearly with the frequency. This simulates the operation of a scalar controlled induction motor in which the V/f ratio is kept constant for flux regulation. The capacitor voltage vector corresponding to the response in Fig. 12(a) is shown in Fig. 12(b). The load current for this case has been chosen as 5 A amplitude. Although the formulation of the system has been done for sinusoidal references for current and voltage, these responses indicate that the control system can successfully cope with slowly changing amplitude and frequency of the reference.

IX. Conclusion

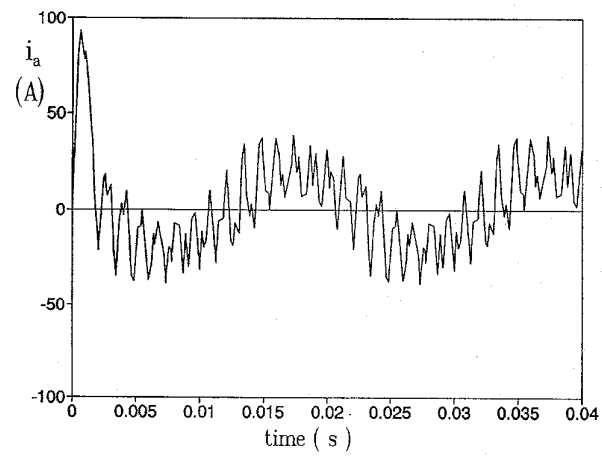
A new deadbeat control strategy for a three-phase inverter with an output LC filter is developed based on space vectors. The mathematical model of the system is seen to be greatly simplified compared with a formulation in the rotating reference frame, rendering the design of the control system easy. It is shown that deadbeat control of output voltage can be achieved in two control steps ideally at no-load or with resistive load, if the computed inverter voltage vector can be exactly generated. However, because of the PWM approximation used to synthesize the demanded voltage vector, a slight error between reference and controlled capacitor voltage results at the end of a sampling interval. The assumption that the load current is constant between sampling instants also introduces some error.



(a)



(b)



(c)

Fig. 11. (a) Inverter current at no-load. (b) Inverter current for the resistive load ($R_0 = 4\ \Omega$). (c) Inverter current for the 20 A (amplitude) load.

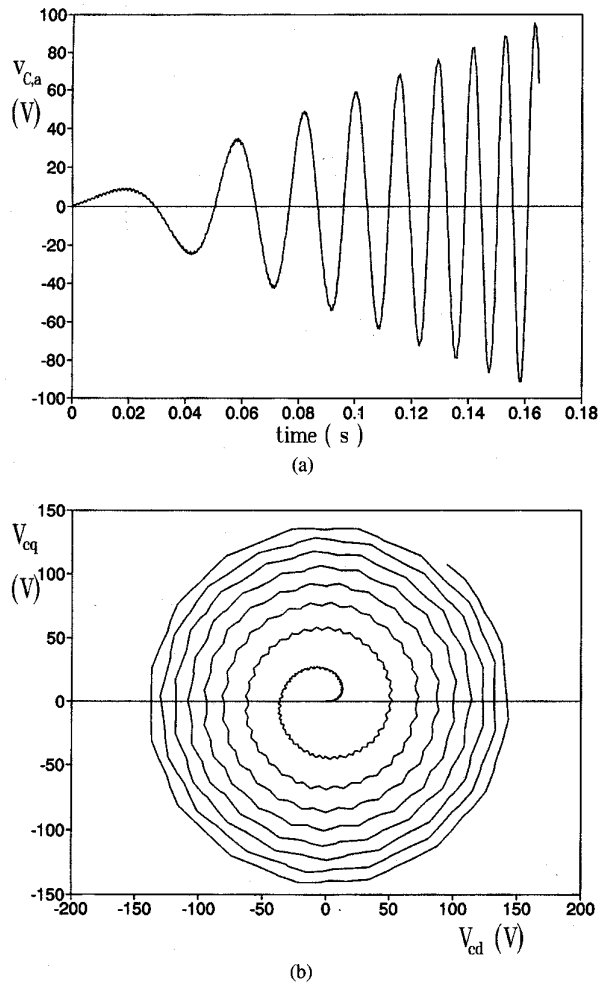


Fig. 12. (a) Phase A capacitor voltage with variable frequency and amplitude reference. (b) Corresponding capacitor voltage vector.

Simulation results indicate that the above errors do not degrade to a significant extent the performance of the control system. The output voltage waveshapes have reasonable distortion in all the linear and nonlinear load cases studied, in spite of the low control sampling rate chosen.

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